

NGUYEN GIA HUY

Backend Engineer

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PROFESSIONAL SUMMARY

Backend engineer with nearly three years of experience building transaction-oriented services and integrations. Integrated a Thought Machine Vault Core banking platform with external services, built a Golang account migration system on AWS (**5x faster**), and built a gRPC, Kafka, MongoDB, and Redis backend serving **1.5K requests/second** at **80–100 ms** with **99.95% uptime**. Works across API contracts, data stores, messaging, and the gateway and identity layers of a **100+ service** microservices ecosystem.

TECHNICAL SKILLS

Languages: Golang, Python, SQL
APIs: gRPC, gRPC-Gateway (REST/JSON), Protocol Buffers, FastAPI
Data Stores: PostgreSQL, MongoDB, Redis, OpenSearch, Elasticsearch, ClickHouse
Messaging & Events: Kafka, RabbitMQ, event-driven services
Service Infrastructure: Kong Gateway, Keycloak (RBAC), Docker, Kubernetes, AWS (ECS, Lambda, EKS, RDS)
Observability: OpenTelemetry, Prometheus, Grafana, ELK

WORK EXPERIENCE

Backend Engineer, Core Banking (Middle Software Engineer) 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

- Integrated **Thought Machine Vault Core** with external backend services, wiring the core ledger into the bank's operational and transaction workflows.
- Shortened account migration from **2.5 hours to 27 minutes (5x)** with a **Golang** migration backend on AWS ECS and Lambda, persisting to **RDS**.
- Implemented Lending and Overdraft transaction logic (interest accrual, penalty and late fees, repayment, settlement) as Python Smart Contracts.
- Gave the transaction logic fast test feedback: simulation tests now finish in **just over 50 seconds instead of 24 minutes** after a Golang rewrite of the Gauge-based framework.

Software Engineer – Backend & Data Services 04/2025 – 03/2026
Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Re-architected data ingestion and indexing while migrating the search and analytics backend from **Elasticsearch to OpenSearch**, improving stability, consistency, and query performance.
- Raised batch throughput by **150%** by reworking data access patterns and running processing steps in parallel.
- Built backend data services on AWS Lambda, ECS, RDS, S3, and Athena for analytics and operational consumers.

Software Engineer – API Gateway & Identity 04/2024 – 04/2025
VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Operated **Kong Gateway** for **100+ microservices**, enforcing service-level and network-level access control.
- Built multi-tenant authentication and authorization on **Keycloak** and RBAC for **20+ customer organizations**.

Blockchain Developer – Event-Driven Backend 06/2023 – 09/2023
EsolLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Built **event-driven backend services** that capture, process, and synchronize on-chain events into off-chain databases and APIs.
- Integrated them with **5+ blockchain networks** (Ethereum, BSC, Polygon, Aptos, Sui), keeping transaction handling consistent and data intact.

SELECTED PROJECTS

DattingQ | Software Engineer *Dating platform*
Built the API layer with **gRPC** services exposed as REST/JSON through **gRPC-Gateway**, backed by **Kafka**, **MongoDB**, and **Redis**.

- Sustained up to **1.5K requests/second** at **80–100 ms** latency, with multi-cluster failover and OpenTelemetry, Prometheus, Grafana, ELK, and ClickHouse observability for **99.95% uptime**.

MeetQ | Software Engineer *AI-powered online meeting platform*
Built a real-time meeting backend on LiveKit integrating OpenAI models for transcription, voice translation, and summaries; designed the real-time translation pipeline.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS) B.Sc. Information Technology – Knowledge Engineering
DevOps on AWS Specialization | Coursera